

Software Requirement Specification

COS 301 - Capstone 2026

Capstone Project: EquityLens

By The Big Five

Field	Detail
Project Name	EquityLens
Client	Amazon Web Services (AWS)
Team	The Big Five (TB5)
Module	COS 301 - Software Engineering
Version	1.0
Last updated	22 May 2026

Name	Student Number
Abdulrahman Sabah(Team Lead)	u24566170
Joshua Heath	u23541475
Antony van Straten	u24590739
Jonty Honey	u23536862

Contents

1. Introduction	4
1.1 Purpose.....	4
1.2 Scope.....	4
1.3 Definitions, Acronyms and Abbreviations	4
2. Project Understanding	6
2.1 Background and Motivation.....	6
2.2 Problem Statement	6
2.3 Proposed Solution.....	6
2.4 Project Rationale.....	6
3. User Stories and User Characteristics	7
3.1 Intended Users	7
3.2 User Stories	7
3.2.1 Authentication	7
3.2.2 Portfolio Management	7
3.2.3 Analytics and Financial Indicators	8
3.2.4 News Correlation	8
3.2.5 AI Interface	8
4. Use Cases and Use Case Diagrams	9
4.1 Use Case Diagram.....	9
4.2 Use Case Descriptions	10
5. Functional Requirements.....	16
5.1 Subsystem 1: Authentication	16
5.2 Subsystem 2: Portfolio Management.....	17
5.3 Subsystem 3: Analytics and Financial Indicators	18
5.4 Subsystem 4: News Correlation	19
5.5 Subsystem 5: AI Interface	20
5.6 Subsystem 6: Dashboard	21
Requirements-Use Case Traceability Matrix.....	22
6. API Service Contracts	23
6.1 Authentication Endpoints	23
6.2 Portfolio Endpoints.....	26

6.3 Analytics Endpoints.....	28
6.4 News Endpoints.....	29
6.5 AI Endpoints	30
6.6 System Endpoints	31
7. Domain Model.....	32
7.1 UML Class Diagram	32
7.2 Entity Descriptions	33
8. Architectural Requirements	36
8.1 Non-Functional Requirements	36
8.2 Architectural Patterns.....	39
8.2.1 Three-Tier Architecture	39
8.2.2 Serverless Architecture	39
8.2.3 Service Layer Pattern	40
8.3 Design Patterns.....	40
8.4 Constraints.....	41
8.4.1 Financial Constraints	41
8.4.2 Security and Compliance Constraints	41
8.4.3 Integration Constraints.....	41
8.4.4 AI Constraints	42
8.4.5 Scalability Constraints	42
9. Technology Requirements	42
9.1 Technology Stack	42
9.2 Development Tools.....	43
9.3 Testing Frameworks.....	44
10 Assumptions and Dependencies	45
10.1 Assumptions.....	45
10.2 Dependencies.....	45

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for EquityLens, an investment analytics platform. It is intended for the development team (The Big Five), project examiners, and stakeholders at AWS to define the technical roadmap and success criteria for this project.

1.2 Scope

EquityLens is a web-based analytics suite for South African retail investors, it covers portfolio ingestion (CSV/PDF/Manual) that provides automated financial indicator calculation, and leverages news-price correlation and a conversational AI interface to benefit the user and financial growth. This project is not orientated around direct stock trading therefore features involving brokerage integration via API (scrapers) or regulated financial advice are not provided.

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
EquityLens	The investment analytics web platform described in this document
SRS	Software Requirements Specification
FR	Functional Requirement
NR	Non-Functional Requirement
UC	Use Case
US	User Story
JWT	JSON Web Token
API	Application Programming Interface
AWS	Amazon Web Services

POPIA	Protection of Personal Information Act (South Africa)
JSE	Johannesburg Stock Exchange
LLM	Large Language Model
ETF	Exchange Traded Fund
CAPM	Capital Asset Pricing Model
RSI	Relative Strength Index
SLA	Service Level Agreement
CI/CD	Continuous Integration / Continuous Deployment

2. Project Understanding

2.1 Background and Motivation

Retail investing in South Africa has surged due to accessible brokerage platforms. However, there is a significant analytical gap between retail investors that have access to the market but lack the institutional-grade tools required to interpret risk metrics and correlate current worldwide events with portfolio volatility.

2.2 Problem Statement

Investors face "data overwhelm". They possess diversified portfolios but cannot easily calculate their total risk exposure or understand why specific stocks are moving in relation to broader news trends without manual, error-prone spreadsheets, preventing investors from reaching their full financial potential.

2.3 Proposed Solution

EquityLens centralizes investment data into a single, AWS-hosted dashboard. It automates the calculation of complex financial indicators and uses a Large Language Model (LLM) to explain these metrics in plain language, bridging the gap between raw data and actionable insight.

2.4 Project Rationale

This system democratizes financial literacy. By providing institutional analytics for free (AWS Free Tier), it empowers users to make more informed decisions, potentially reducing financial loss due to lack of diversified risk awareness.

3. User Stories and User Characteristics

3.1 Intended Users

User Type	Description
Retail Investor	Primary user - imports portfolio data and uses analytics features
Administrator	Manages system configuration and user accounts

3.2 User Stories

Format: As a [user type], I want to [action] so that [benefit].

3.2.1 Authentication

ID	User Story
US-1	As a new user, I want to create an account so that my portfolio data is saved securely.
US-2	As a returning user, I want to log in using JWT-based security so that my session remains active while I work
US-3	As a security-conscious user, I want to log out so that others cannot access my data on a shared device.

3.2.2 Portfolio Management

ID	User Story
US-4	As an investor, I want to upload a CSV of my holdings so that I don't have to enter them manually.
US-5	As an EasyEquities user, I want to upload my PDF statement so the system can parse my holdings automatically.
US-6	As a user with a small portfolio, I want to manually add a ticker so I can track new investments immediately.

3.2.3 Analytics and Financial Indicators

ID	User Story
US-7	As an investor, I want to see my portfolio's Sharpe Ratio so that I can judge if my returns justify the risk.
US-8	As a user, I want tooltips for every indicator so I can learn what the numbers mean.
US-9	As a technical user, I want to see the RSI for my stocks to identify overbought or oversold conditions.

3.2.4 News Correlation

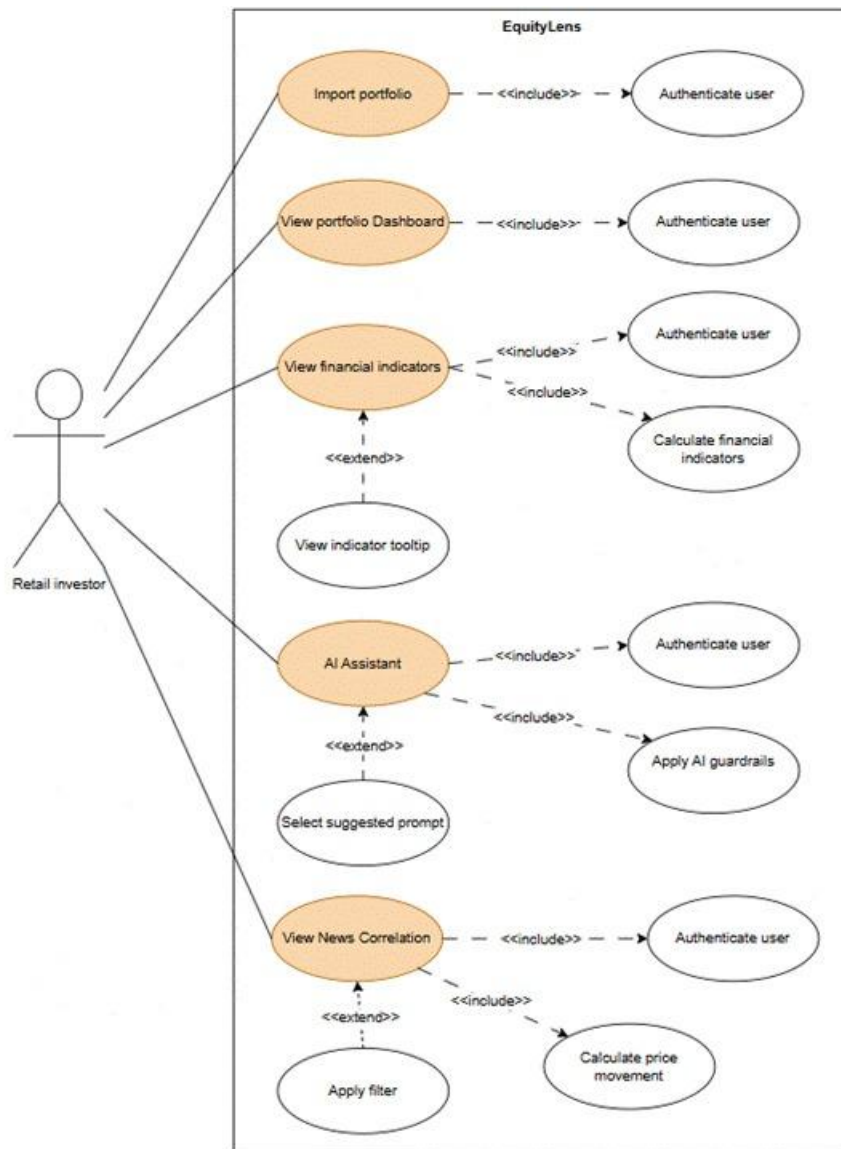
ID	User Story
US-10	As a user, I want to see news headlines next to my biggest losers so I can understand the drop.
US-11	As an investor, I want news filtered by the companies I actually own to save time.

3.2.5 AI Interface

ID	User Story
US-12	As a novice, I want to ask "Is my portfolio diversified?" to get a natural language summary.
US-13	As a user, I want the AI to explain the CAPM of my portfolio so I understand my market sensitivity.

4. Use Cases and Use Case Diagrams

4.1 Use Case Diagram



4.2 Use Case Descriptions

UC-1: Import Portfolio

Use Case ID	UC-1
Use Case Name	Import Portfolio
Actor(s)	Retail investor
Description	The investor uploads existing holdings by CSV, PDF statement or by hand (fallback) into EquityLens. The system validates inputs and displays parsed holdings in a holdings table.
Preconditions	The user is on the Portfolio page. They have either a portfolio file in CSV or PDF format, or the data needed to enter holdings manually.
Postconditions	Portfolio holdings are shown in the holdings table on the portfolio page
Basic Flow	1. Clicking import opens the upload window 2. User selects CSV file from device 3. Backend parses file and extracts the content it needs to find 4. Backend validates the content is correct 5. System then saves the holdings and frontend displays them in a table
Alternative Flow	1. Upload a pdf: Instead of selecting a CSV file, the user selects a PDF at step 2. 2. Manual entry(The fallback): At step 1 the user opens an entry form instead of clicking import and enters the content that is needed (replaces step 2 and 3 too, joins again at step 4).
Exception Flow	1. Malformed file: When validation step (step 4) fails. The system rejects the upload and shows an error message. Prompts/Allows the user to retry by keeping the upload window open. 2. No content: The backend parsing succeeds but no entries were found. The system outputs an error message that

	nothing was found. Prompts the user to enter manually or upload a different file.
--	---

UC-2: Portfolio Dashboard

Use Case ID	UC-2
Use Case Name	Portfolio Dashboard
Actor(s)	Retail investor
Description	The investor opens the dashboard page that summarises their portfolio. This dashboard contains: a sector allocation pie chart, a performance line chart over time and a holdings table showing ticker, quantity, current value and gain/loss.
Preconditions	The user is authenticated and the user has at least one holding in their portfolio
Postconditions	The dashboard page shows the sector pie chart, performance line chart and holdings table for user uploaded portfolio.
Basic Flow	1. User goes to dashboard page 2. Frontend verifies users session and system verifies/confirms authentication 3. The backend returns the holdings, sector breakdown, and performance time series. 4. The Dashboard displays the pie chart (Recharts), performance line chart, summary card, and holdings table.
Alternative Flow	No/Empty portfolio: At step 3 the backend returns an empty holdings list so the dashboard shows an empty state with a button prompting the user to import their portfolio.

Exception Flow	1. Unauthenticated user: System redirects user to login/register page if the session is invalid/missing. Dashboard not displayed 2. Backend issue: At step 3 the backend returns an error, dashboard displays error state with retry prompt. Dashboard does not return tables and charts
-----------------------	---

UC-3: Financial Indicator View

Use Case ID	UC-3
Use Case Name	Financial Indicator view
Actor(s)	Retail Investor
Description	The investor opens the analytics page to see key financial indicators which have been calculated from their portfolio. Each indicator is represented as a card with its name, value and a plain-language tooltip explanation so the user can make judgments without needing finance training. Financial indicators are: Altman Z-score, P/E ratio, Sortino Ratio, CAPM ratio, Sharpe ratio, RSI and beta value.
Preconditions	The user is authenticated and has holdings in their portfolio. There must also be enough price data to calculate each indicator's results.
Postconditions	The analytics page displays an indicator card for each indicator which shows its value and name.
Basic Flow	1. User goes to analytics page 2. Frontend requests calculations from the backend 3. Backend gathers users saved holdings 4. System calculates financial indicators of all stocks that the user is currently holding. 5. Backend then sends both of these calculations to frontend 6. Frontend then displays those cards with names, values and tooltips

Alternative Flow	View Tooltip: When a user hovers over an indicator name, a clear text explanation is displayed and moving the cursor away hides the message and does not affect the page.
Exception Flow	1. Insufficient Data: At step 4 or 5, the indicator cannot be calculated. The card for that indicator renders an N/A state with a short explanation. Other indicator cards still render normally. 2. Calculation failure: One of the calculation modules fail. The affected card displays an error state with a retry prompt. The rest of the analytics page is still usable.

UC-4: AI Assistant

Use Case ID	UC-4
Use Case Name	AI Assistant
Actor(s)	Retail Investor
Description	The investor submits a natural-language question about their portfolio to the AI assistant. The system processes the query and returns an analysis-based response. The assistant will not provide explicit buy or sell recommendations - this guardrail is enforced at the system level.
Preconditions	The user is authenticated and on the AI Chat page.
Postconditions	A response is displayed in the chat interface. The conversation history is visible in the session.
Basic Flow	1. User types a question into the chat input field 2. User submits the message 3. System sends the query along with the user's portfolio context to the AI layer 4. AI layer processes the query and generates a response 5. Response is displayed in the chat interface below the user's message

Alternative Flow	Suggested prompt selected: Instead of typing, the user clicks one of the suggested prompt chips shown on load. The chip pre-fills the input field and the flow continues from step 2.
Exception Flow	1. Guardrail triggered: The user asks for an explicit buy or sell recommendation. The system detects the intent and returns a response explaining that EquityLens provides analysis and information only. The user is redirected to the financial indicators page for data to inform their own decision. 2. AI service unavailable: The primary AI layer fails to respond. The system automatically falls back to the secondary layer without any user-visible error. If both are unavailable the system displays an error message and prompts the user to try again.

UC-5: View News Correlation

Use Case ID	UC-5
Use Case Name	View News Correlation
Actor(s)	Retail Investor
Description	The investor views financial news articles correlated to their portfolio holdings. Articles are filterable by ticker and date range. Each article displays a price movement indicator showing how the related stock moved around the time of publication.
Preconditions	The user is authenticated and on the News page.
Postconditions	A list of news articles relevant to the user's holdings is displayed, filterable by ticker and date.
Basic Flow	1. User navigates to the News page 2. System loads articles relevant to all holdings in the user's portfolio by default 3. Articles are displayed with headline, source, publication time, related ticker badge, and price movement indicator 4. User clicks an article to open the full article in a new tab at the source publication

Alternative Flow	1. Filter by ticker: The user clicks a ticker chip above the article list. The system filters the displayed articles to show only those correlated to that specific holding. 2. Filter by date range: The user selects a date range from the date filter dropdown. The system filters articles to the selected window - Today, This Week, This Month, or a custom range.
Exception Flow	1. No articles found: The system finds no articles for the selected ticker or date range. An empty state message is displayed prompting the user to broaden their filter or check back later. 2. News API unavailable: The external news service fails to respond. The system displays an error message indicating news is temporarily unavailable and prompts the user to try again.

5. Functional Requirements

5.1 Subsystem 1: Authentication

FR-1: User Registration Management

- FR-1.1: The system shall allow users to register with a full name, valid email address, and password
- FR-1.2: The system shall validate email format before submission
- FR-1.3: The system shall enforce a minimum password length of 8 characters containing at least one uppercase letter, one lowercase letter, one number, and one special character
- FR-1.4: The system shall display clear inline error messages for invalid or missing registration inputs
- FR-1.5: The system shall reject registration attempts using an email address already associated with an existing account
- FR-1.6: The system shall redirect users to the login page upon successful registration.

FR-2: User Authentication & Session Management

- FR-2.1: The system shall authenticate users via email and password
- FR-2.2: The system shall issue a JWT access token upon successful login
- FR-2.3: The system shall store the JWT token securely on the client side under a namespaced key
- FR-2.4: The system shall include the JWT token as a Bearer token in the Authorization header of all protected API requests
- FR-2.5: The system shall redirect unauthenticated users attempting to access protected routes to the login page
- FR-2.6: The system shall display an appropriate error message when invalid credentials are provided
- FR-2.7: The system shall allow users to log out, removing the active session token from client storage

5.2 Subsystem 2: Portfolio Management

FR-3: Automated Data Ingestion

- FR-3.1: The system shall allow users to import portfolio data via CSV file upload
- FR-3.2: The system shall allow users to import portfolio data via PDF brokerage statement upload using pdfPlumber to extract holdings automatically
- FR-3.3: The system shall validate uploaded files and reject unsupported formats with a clear error message
- FR-3.4: The system shall display a preview of parsed holdings for user confirmation before saving to the database
- FR-3.5: The system shall associate all imported portfolio data with the authenticated user account

FR-4: Manual Data Management

- FR-4.1: The system shall allow users to manually enter portfolio holdings including ticker symbol, quantity, and purchase price
- FR-4.2: The system shall validate manually entered ticker symbols against known JSE-listed stocks

FR-5: Holdings Display

- FR-5.1: The system shall display imported holdings in a structured table showing ticker symbol, company name, sector, quantity, average purchase price, current price, total value, profit/loss in Rand, and profit/loss as a percentage
- FR-5.2: The system shall display positive profit/loss values in green and negative values in red
- FR-5.3: The system shall ensure portfolio data is scoped to the authenticated user and never visible to any other user

5.3 Subsystem 3: Analytics and Financial Indicators

FR-6: Portfolio Visualisation

- FR-6.1: The system shall display a sector allocation pie chart based on the user's imported holdings
- FR-6.2: The system shall display portfolio performance over time as a line chart
- FR-6.3: The system shall compare portfolio performance against the JSE All Share Index as a benchmark
- FR-6.4: The system shall display a dividend income bar chart showing monthly income from holdings

FR-7: Financial Indicator Calculation

- FR-7.1: The system shall calculate and display the Sharpe Ratio as portfolio return minus risk-free rate divided by portfolio standard deviation
- FR-7.2: The system shall calculate and display the P/E Ratio for individual holdings
- FR-7.3: The system shall calculate and display the CAPM expected return for individual holdings
- FR-7.4: The system shall calculate and display the Altman Z-Score for individual holdings, classifying results as safe (>2.99), grey zone ($1.81-2.99$), or distress (<1.81)
- FR-7.5: The system shall calculate and display Beta for individual holdings relative to the JSE All Share benchmark
- FR-7.6: The system shall calculate and display the RSI over a 14-day period, classifying results as overbought (>70), neutral ($30-70$), or oversold (<30)
- FR-7.7: The system shall calculate and display the Sortino Ratio using only downside deviation in the denominator

FR-8: Indicator Presentation

- FR-8.1: The system shall provide plain-language tooltips for each financial indicator explaining its meaning and interpretation

- FR-8.2: The system shall display an N/A state with an explanation when insufficient data exists to calculate an indicator
- FR-8.3: The system shall colour-code indicator values as positive, neutral, or negative based on predefined thresholds

5.4 Subsystem 4: News Correlation

FR-9: Price Movement Detection

- FR-9.1: The system shall retrieve market price data for all holdings in the user's portfolio
- FR-9.2: The system shall detect significant price movements exceeding a defined volatility threshold

FR-10: News Retrieval and Correlation

- FR-10.1: The system shall automatically query NewsAPI.org for financial news articles published within a 48-hour window of a detected price movement
- FR-10.2: The system shall filter retrieved articles to those relevant to the specific holding's ticker symbol or company name
- FR-10.3: The system shall display correlated news articles alongside the corresponding price movement with headline, source, publication time, and related ticker badge
- FR-10.4: The system shall allow users to filter displayed news articles by ticker symbol and date range
- FR-10.5: The system shall clearly indicate when news data is unavailable for a given holding

5.5 Subsystem 5: AI Interface

FR-11: Context-Aware Query Engine

- FR-11.1: The system shall provide a conversational chat interface allowing users to submit natural-language questions about their portfolio
- FR-11.2: The system shall build prompt context by aggregating the user's portfolio holdings, calculated financial indicators, and relevant market data before querying the AI model
- FR-11.3: The system shall display suggested prompt chips on load to guide users toward supported query types
- FR-11.4: The system shall maintain conversation history within the active session

FR-12: Compliance and Guardrails

- FR-12.1: The system shall screen all AI responses against a financial safety system prompt that prevents explicit buy or sell recommendations
- FR-12.2: The system shall detect queries requesting explicit investment advice and respond with a disclaimer redirecting the user to the Analytics page
- FR-12.3: The system shall append a disclaimer to all AI responses clarifying that responses are informational and not regulated financial advice
- FR-12.4: The guardrail implementation shall be documented in this specification

FR-13: AI Service Resilience

- FR-13.1: The system shall provide a fallback response mechanism in the event the primary AI service is unavailable
- FR-13.2: The fallback mechanism shall activate transparently with no user-visible error message

5.6 Subsystem 6: Dashboard

FR-14: Dashboard Layout and Content

- FR-14.1: The system shall provide a central dashboard displaying the user's top four holdings as summary cards showing ticker symbol, current price, and percentage change
- FR-14.2: The system shall display a portfolio performance line chart on the dashboard with toggles for monthly, quarterly, and annual views
- FR-14.3: The system shall display a watchlist on the dashboard showing all holdings with current price and daily percentage change
- FR-14.4: The system shall display a full holdings table on the dashboard with all position details

FR-15: Dashboard Interactivity

- FR-15.1: The system shall allow users to toggle between light and dark display themes
- FR-15.2: The system shall display a live market data ticker in the topbar showing JSE All Share Index, USD/ZAR exchange rate, and Brent crude price

Requirements-Use Case Traceability Matrix

Requirements	Priory Weight	UC1 Import Portfolio	UC2 Portfolio Dashbored	UC1 Financial Indiactors	UC1 AI Assistant	UC1 News Correlation
FR-1 Registration Management	3		X	X	X	X
FR-2 Authentication & Session	3	X	X	X	X	X
FR-3 Automated Data Ingestion	3	X				
FR-4 Manual Data Management	3	X				
FR-5 Holdings Display	3	X	X			
FR-6 Portfolio Visualization	3		X			
FR-7 Financial Indicator Calculation	2			X	X	
FR-8 Indicator Presentation	2			X	X	
FR-9 Price Movement Detection	2					X
FR-10 News Retrieval & Correlation	3					X
FR-11 Context-Aware Query Engine	3				X	

FR-12 Compliance & Guardrails	3				X	
FR-13 AI Service Resilience	2				X	
FR-14 Dashboard Layout & Content	3		X			
FR-15 Dashboard Interactivity	2		X			
Score		12	17	10	18	11

6. API Service Contracts

6.1 Authentication Endpoints

POST /api/auth/register

Method	POST
Path	/api/auth/register
Description	Register a new user account with full name, email and password
Authentication	Not required

Request Body:

```
{
  "full_name": "Example User",
  "email": "example@example.com",
  "password": "Password1!"
}
```

Response - 201 Created:

```
{  
  "id": "uuid",  
  "email": "example@example.com",  
  "full_name": "Example User",  
  "is_active": true,  
  "created_at": "2026-05-17T12:00:00Z"  
}
```

Error Responses:

Status Code	Description
409	A user with this email address already exists
422	Validation failed - invalid email format, password too weak, or missing required fields

POST /api/auth/login

Method	POST
Path	/api/auth/login
Description	Authenticate a user and return a JWT access token
Authentication	Not required

Request Body:

```
{  
  "email": "example@example.com",  
  "password": "Password1!"  
}
```



```
}
```

Response - 200 OK:

```
{  
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",  
  "token_type": "bearer"  
}
```

Error Responses:

Status Code	Description
401	Invalid credentials - email not found or password incorrect
422	Validation failed - malformed request body

GET /api/auth/me

Method	GET
Path	/api/auth/me
Description	Return the currently authenticated user's profile
Authentication	Required - Bearer token in Authorization header

Response - 200 OK:

```
{  
  "id": "uuid",  
  "email": "example@example.com",  
  "full_name": "Example User",  
  "is_active": true,
```

```
"created_at": "2026-05-17T12:00:00Z"
}
```

Error Responses:

Status Code	Description
401	Missing, expired, or invalid Bearer token

6.2 Portfolio Endpoints

GET /api/portfolio

Method	GET
Path	/api/portfolio
Description	Return the authenticated user's full portfolio including summary, holdings, sector allocation and performance history
Authentication	Required - Bearer token

Response - 200 OK:

```
{
  "summary": {
    "total_value": 125000.00,
    "total_gain_loss": 12500.00,
    "total_gain_loss_pct": 11.11
  },
  "holdings": [
    {
```

```

    "ticker": "NPN",
    "name": "Naspers",
    "sector": "Technology",
    "quantity": 10,
    "avg_price": 2800.00,
    "current_price": 3150.00,
    "value": 31500.00,
    "gain_loss": 3500.00,
    "gain_loss_pct": 12.50
  }
],
"sectors": [
  { "sector": "Technology", "allocation_pct": 45.2 }
],
"performance": [
  { "date": "2026-01-01", "value": 110000.00 }
]
}

```

GET /api/portfolio/summary

Method	GET
Path	/api/portfolio/summary
Description	Return portfolio summary statistics only
Authentication	Required - Bearer token

GET /api/portfolio/sectors

Method	GET
Path	/api/portfolio/sectors
Description	Return sector allocation breakdown for the user's portfolio
Authentication	Required - Bearer token

GET /api/portfolio/performance

Method	GET
Path	/api/portfolio/performance
Description	Return portfolio value history over time
Authentication	Required - Bearer token

6.3 Analytics Endpoints

GET /api/indicators/{ticker}

Method	GET
Path	/api/indicators/{ticker}
Description	Return calculated financial indicators for the specified ticker symbol
Authentication	Required - Bearer token

Response - 200 OK:

```
{  
  "ticker": "NPN",  
  "pe_ratio": 28.4,
```

```
"sharpe_ratio": 1.32,  
"beta": 0.87,  
"rsi": 54.2,  
"altman_z_score": 3.1,  
"sortino_ratio": 1.75,  
"capm_expected_return": 0.112  
}
```

Status Code	Description
401	Unauthorised
404	Ticker not found or no data available

6.4 News Endpoints

GET /api/news/{ticker}

Method	GET
Path	/api/news/{ticker}
Description	Return financial news articles correlated to the specified ticker and its price movements
Authentication	Required - Bearer token

Response - 200 OK:

```
{  
  "ticker": "NPN",  
  "articles": [  
    {  
      "headline": "Naspers reports strong Q4 results",  
      "summary": "Naspers reports strong Q4 results",  
      "url": "https://www.naspers.com/~/media/press-releases/2020/12/20201201-naspers-q4-2020-report.pdf",  
      "published": "2020-12-01T14:00:00Z",  
      "author": "Naspers",  
      "category": "Naspers",  
      "tags": ["Naspers", "Q4", "2020", "results"]  
    }  
  ]  
}
```

```

    "source": "BusinessDay",
    "url": "https://businessday.co.za/...",
    "published_at": "2026-05-16T08:00:00Z"
  }
]
}

```

Status Code	Description
401	Unauthorised
404	No news found for the specified ticker

6.5 AI Endpoints

POST /api/ai/chat

Method	POST
Path	/api/ai/chat
Description	Submit a natural-language query about the user's portfolio and receive an AI-generated response with guardrails applied
Authentication	Required - Bearer token

Request Body:

```

{
  "message": "How is my portfolio performing compared to the JSE All Share?"
}

```

Response - 200 OK:

```

{

```

"response": "Based on your current holdings, your portfolio has returned 11.1% year-to-date, compared to the JSE All Share benchmark return of 8.3% over the same period.",

"model_used": "aws-bedrock"

}

Status Code	Description
401	Unauthorised
422	Empty or malformed message body

6.6 System Endpoints

GET /health

Method	GET
Path	/health
Description	Health check endpoint - returns system status
Authentication	Not required

Response - 200 OK:

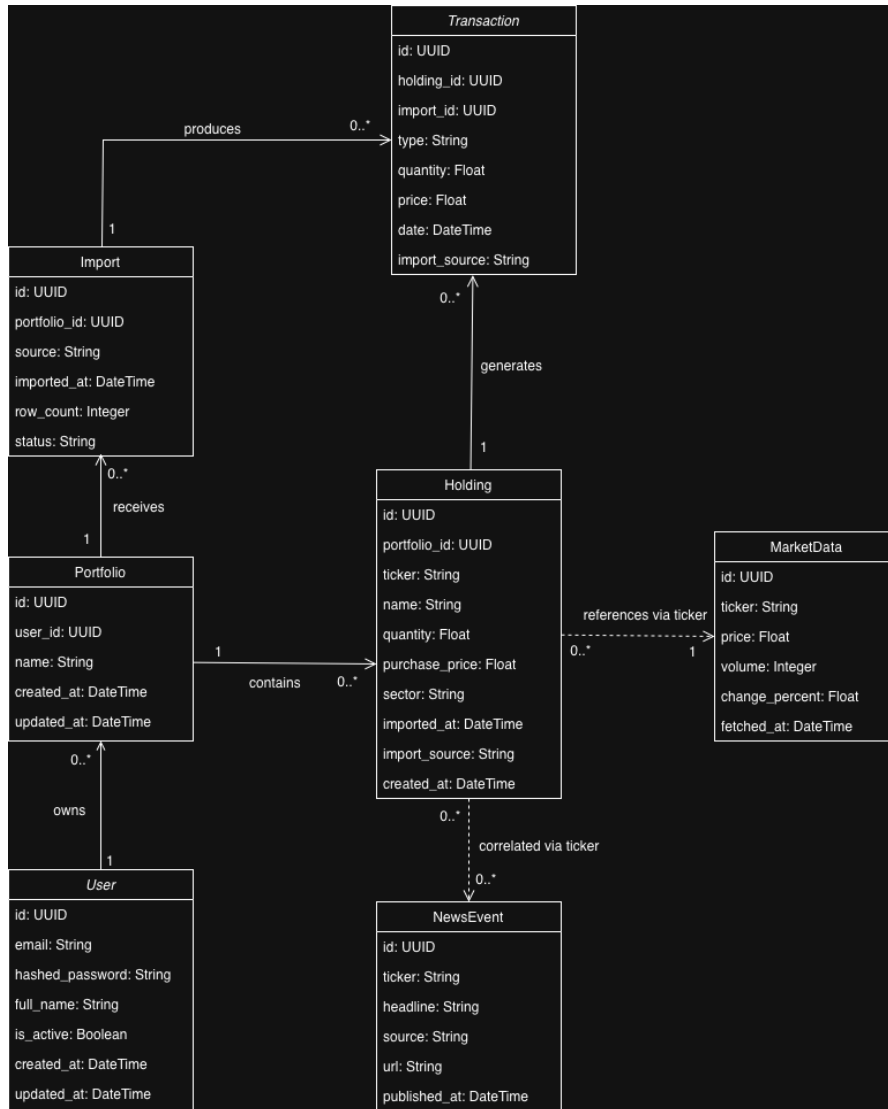
{

"status": "ok"

}

7. Domain Model

7.1 UML Class Diagram



7.2 Entity Descriptions

User

Attribute	Type	Description
id	UUID	Unique identifier - primary key
email	String	User email address - unique and indexed
hashed_password	String	Bcrypt hashed password - never stored in plain text
full_name	String	User's full name
is_active	Boolean	Whether the account is active
created_at	DateTime	Account creation timestamp
updated_at	DateTime	Last update timestamp

Portfolio

Attribute	Type	Description
id	UUID	Unique identifier - primary key
user_id	UUID	Foreign key reference to User
name	String	Portfolio name
created_at	DateTime	Creation timestamp
updated_at	DateTime	Last modification timestamp

Holding

Attribute	Type	Description
id	UUID	Unique identifier - primary key
portfolio_id	UUID	Foreign key reference to Portfolio

ticker	String	Stock ticker symbol (e.g. NPN, MTN)
name	String	Company name
quantity	Float	Number of shares held
purchase_price	Float	Average purchase price per share
sector	String	Industry sector classification
imported_at	DateTime	Timestamp of the last import that modified this holding
import_source	String	Source of the last update. E.g. PDF, CSV or Manual
created_at	DateTime	Record creation timestamp

Transaction

Attribute	Type	Description
id	UUID	Unique identifier - primary key
holding_id	UUID	Foreign key reference to Holding
import_id	UUID	Foreign key reference to Import
type	String	Transaction type - BUY or SELL
quantity	Float	Number of shares in transaction
price	Float	Price per share at time of transaction
date	DateTime	Date and time of transaction
import_source	String	Ingestion method

MarketData

Attribute	Type	Description
id	UUID	Unique identifier - primary key
ticker	String	Stock ticker symbol

price	Float	Current or historical price
volume	Integer	Trading volume
change_percent	Float	Price change since previous close
fetches_at	DateTime	Timestamp when data was last fetched - used for caching

NewsEvent

Attribute	Type	Description
id	UUID	Unique identifier - primary key
ticker	String	Related stock ticker symbol
headline	String	News article headline
source	String	Publication source
url	String	Direct URL to article
published_at	DateTime	Article publication date and time

Import

Attribute	Type	Description
id	UUID	Unique identifier - primary key
portfolio_id	String	Related stock ticker symbol
source	String	Publication source
imported_at	DateTime	Timestamp of the import event
row_count	Integer	Number of holding rows processed
status	String	Outcome of the import. SUCCESS - FAILED - PARTIAL

8. Architectural Requirements

8.1 Non-Functional Requirements

NR-1: Security

- NR-1.1: All API endpoints except /health, /register, and /login shall require a valid JWT Bearer token - unauthenticated requests shall receive a 401 response
- NR-1.2: All passwords shall be hashed using bcrypt with a minimum cost factor of 12 - plain text passwords shall never be stored
- NR-1.3: JWT access tokens shall expire after 30 minutes to limit exposure from token theft
- NR-1.4: All user portfolio data shall be scoped to the authenticated user at the database query level - no user shall be able to access another user's data
- NR-1.5: All data in transit shall be encrypted using TLS
- NR-1.6: All data at rest shall be encrypted using AES-256
- NR-1.7: Raw brokerage credentials shall never be stored in the system - portfolio data may only be imported via file upload or manual entry
- NR-1.8: All user data handling shall be fully compliant with the POPIA Act

NR-2: Performance

- NR-2.1: API endpoints shall respond within 500ms at the 95th percentile under normal load
- NR-2.2: Frontend initial page load shall complete within 3 seconds on a standard broadband connection
- NR-2.3: Dashboard visualisations shall render within 3 seconds for portfolios containing up to 50 holdings
- NR-2.4: Market data shall be cached in PostgreSQL and only re-fetched if the cached data is older than 15 minutes

NR-3: Reliability

- NR-3.1: The system shall maintain at least 99% uptime during normal operating hours measured via Uptime Robot
- NR-3.2: The AI fallback mechanism shall activate within 2 seconds of primary service failure with no user-visible error message
- NR-3.3: The CI pipeline shall prevent broken code from reaching the main branch by blocking merges on any test failure

NR-4: Scalability

- NR-4.1: The backend shall support a minimum of 50 concurrent users without performance degradation via AWS Lambda auto-scaling
- NR-4.2: The database shall support growing volumes of historical market data with p95 query times remaining below 200ms up to 1 million rows
- NR-4.3: Horizontal scaling shall be achievable without significant architectural changes

NR-5: Maintainability

- NR-5.1: All new code shall maintain a minimum of 70% test coverage enforced via Codecov on every pull request to dev and main
- NR-5.2: The CI pipeline shall run the full frontend and backend test suites automatically on every push and pull request
- NR-5.3: All version control operations shall be performed via the command line using system Git as per project requirements
- NR-5.4: All infrastructure shall be defined as code using AWS SAM to ensure reproducible deployments

NR-6: Accessibility and Usability

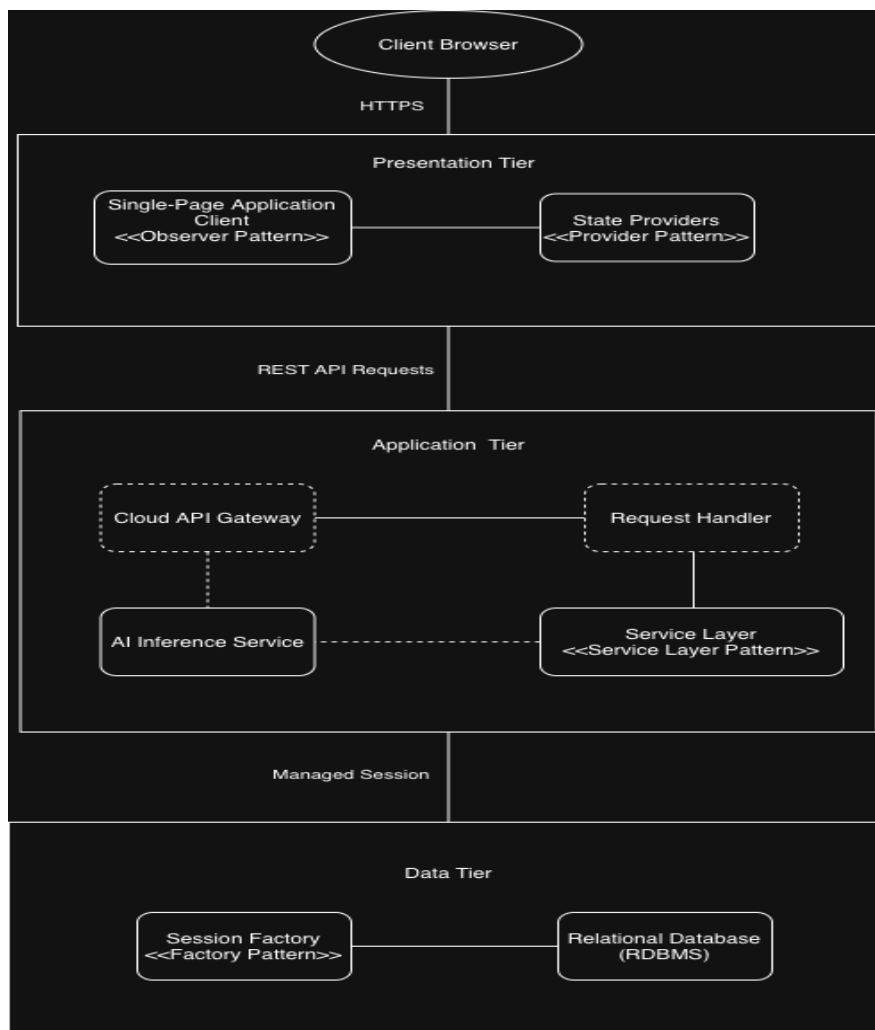
- NR-6.1: The frontend shall meet WCAG 2.1 AA accessibility standards verified via aria-label attributes, keyboard navigability, and sufficient colour contrast
- NR-6.2: All interactive elements shall be keyboard navigable with clearly visible focus states
- NR-6.3: All charts shall provide a text alternative or data table for screen reader compatibility

- NR-6.4: The interface shall be fully responsive across desktop (1280px+), tablet (768px–1279px), and mobile (320px–767px) with no horizontal scrolling on mobile
- NR-6.5: The interface shall respect the prefers-reduced-motion media query and disable animations for users who prefer reduced motion

NR-7: AI Compliance

- NR-7.1: The conversational AI interface shall implement documented guardrails at the system prompt level preventing explicit buy or sell recommendations
- NR-7.2: All AI responses shall include a disclaimer clarifying that responses are informational and not regulated financial advice
- NR-7.3: AI capabilities shall use only free-tier or open-source model APIs to comply with infrastructure cost constraints

8.2 Architectural Patterns



8.2.1 Three-Tier Architecture

EquityLens follows a three-tier architecture. The presentation tier is a React + Vite single-page application hosted on AWS Amplify, responsible for rendering UI and handling user interaction. The application tier is a FastAPI backend deployed on AWS Lambda via API Gateway which contains and handles all business logic, authentication, and data processing. The data tier is an AWS RDS PostgreSQL instance storing user accounts, portfolio data, and cached market data. The three tiers communicate via HTTP REST APIs with no direct database access from the frontend.

8.2.2 Serverless Architecture

The backend is deployed as a serverless application using AWS Lambda and AWS API Gateway. Lambda functions execute on demand with no persistent servers to provision or manage and scale automatically with incoming traffic. This keeps the system within

AWS Free Tier limits during development and early usage while still providing high-end infrastructure.

8.2.3 Service Layer Pattern

Business logic is encapsulated in dedicated service classes under `backend/app/services/`. Routers in `backend/app/routers/` handle only HTTP concerns (request parsing, response formatting, status codes). This separation keeps routers testable and ensures business logic can be unit tested independently of the HTTP layer.

8.3 Design Patterns

Pattern	Location in System	Rationale
Service Layer Pattern	<code>backend/app/services/</code>	Separates business logic from HTTP routing, improving testability and maintainability
Provider Pattern	<code>frontend/src/context/</code>	React context providers (<code>AuthContext</code> , <code>ThemeContext</code>) make global state available without prop drilling
Observer Pattern	React state and context updates	Component re-renders are driven by state changes, implementing an implicit observer model across the UI
Strategy Pattern	API fallback mechanism	The primary and fallback LLM implementations are interchangeable strategies selected at runtime
Factory Pattern	Database session management	<code>get_db()</code> in <code>app/database.py</code> acts as a factory, yielding and closing SQLAlchemy sessions consistently across all request handlers

8.4 Constraints

8.4.1 Financial Constraints

ID	Constraint
C-1	All infrastructure must operate within AWS Free Tier limits at zero cost
C-2	All non-AWS tools must be open-source or free-tier
C-3	AI capabilities must leverage free-tier LLM APIs exclusively

8.4.2 Security and Compliance Constraints

ID	Constraint
C-4	All data must be encrypted in transit (TLS) and at rest (AES-256)
C-5	All user data handling must be fully compliant with the POPIA Act
C-6	Raw brokerage credentials must never be stored in the system
C-7	Portfolio data may only be imported via file upload or manual entry
C-8	User portfolio data must not be visible to any other user

8.4.3 Integration Constraints

ID	Constraint
C-9	Market data fetching must respect API rate limits with appropriate PostgreSQL timestamp-based caching
C-10	The platform must support at least one South African brokerage for portfolio PDF import

8.4.4 AI Constraints

ID	Constraint
C-11	The conversational AI interface must include documented guardrails preventing explicit buy or sell recommendations
C-12	Guardrails must be implemented at the system prompt level and documented in this specification

8.4.5 Scalability Constraints

ID	Constraint
C-13	The architecture must support multiple concurrent users and growing volumes of historical market data
C-14	Horizontal scaling must be achievable without significant architectural changes

9. Technology Requirements

9.1 Technology Stack

The following table represents the full production technology stack. Current local development uses Docker Compose with local PostgreSQL and JWT-based authentication in place of AWS RDS and Cognito respectively.

Category	Technology
Frontend Framework	React + Vite
UI Styling	TailwindCSS
Data Visualisation	Recharts
Frontend Hosting	AWS Amplify

Backend Framework	FastAPI
Serverless Compute	AWS Lambda
API Management	AWS API Gateway
Authentication	AWS Cognito
Database	AWS RDS PostgreSQL
File Storage	AWS S3
AI - Primary	AWS Bedrock
Market Data - Primary	yfinance
Market Data - Fallback	Alpha Vantage
News Correlation	Finnhub.io + MarketAux
PDF Parsing	pdfPlumber
Financial Computation	pandas + numpy + scipy
CI/CD	GitHub Actions
Infrastructure as Code	AWS SAM

9.2 Development Tools

Tool	Purpose
Git (command line only)	Version control - all commits via CLI as per project requirements
GitHub	Repository hosting, issue tracking, project board management
Docker + Docker Compose	Local development environment - orchestrates frontend, backend and database
Postman	API testing and endpoint documentation
PDF Wireframes	Wireframe and UI design

VS Code	Primary IDE
---------	-------------

9.3 Testing Frameworks

Framework	Layer	Purpose
Vitest	Frontend	Unit and component testing
React Testing Library	Frontend	Component behaviour and accessibility testing
pytest	Backend	Unit and integration testing
pytest-cov	Backend	Code coverage measurement and reporting
Codecov	CI/CD	Coverage badge and trend reporting on GitHub

10 Assumptions and Dependencies

10.1 Assumptions

ID	Assumption
A-1	Users have a basic understanding of stock ticker symbols and financial terminology
A-2	Users have an existing investment portfolio with at least one holding
A-3	Users have access to a modern web browser with JavaScript enabled
A-4	Portfolio PDF statements are sourced from supported South African brokerages such as EasyEquities
A-5	Market data provided by yfinance accurately reflects JSE-listed stock prices

10.2 Dependencies

ID	Dependency	Assumption
D-1	yfinance Python library	Market data unavailable - Alpha Vantage fallback activates
D-2	MarketAux API	News correlation feature unavailable
D-3	AWS Bedrock	AI assistant unavailable - rule-based keyword responses
D-4	AWS Free Tier	Infrastructure costs increase - violates constraint C-1
D-5	GitHub Actions	CI pipeline unavailable - manual testing required
D-6	pdfPlumber library	PDF portfolio import unavailable - CSV and manual entry still functional